

BOSTON UNIVERSITY
ELECTRICAL & COMPUTER ENGINEERING

FALL 2026
OSAMA ALSHAYKH

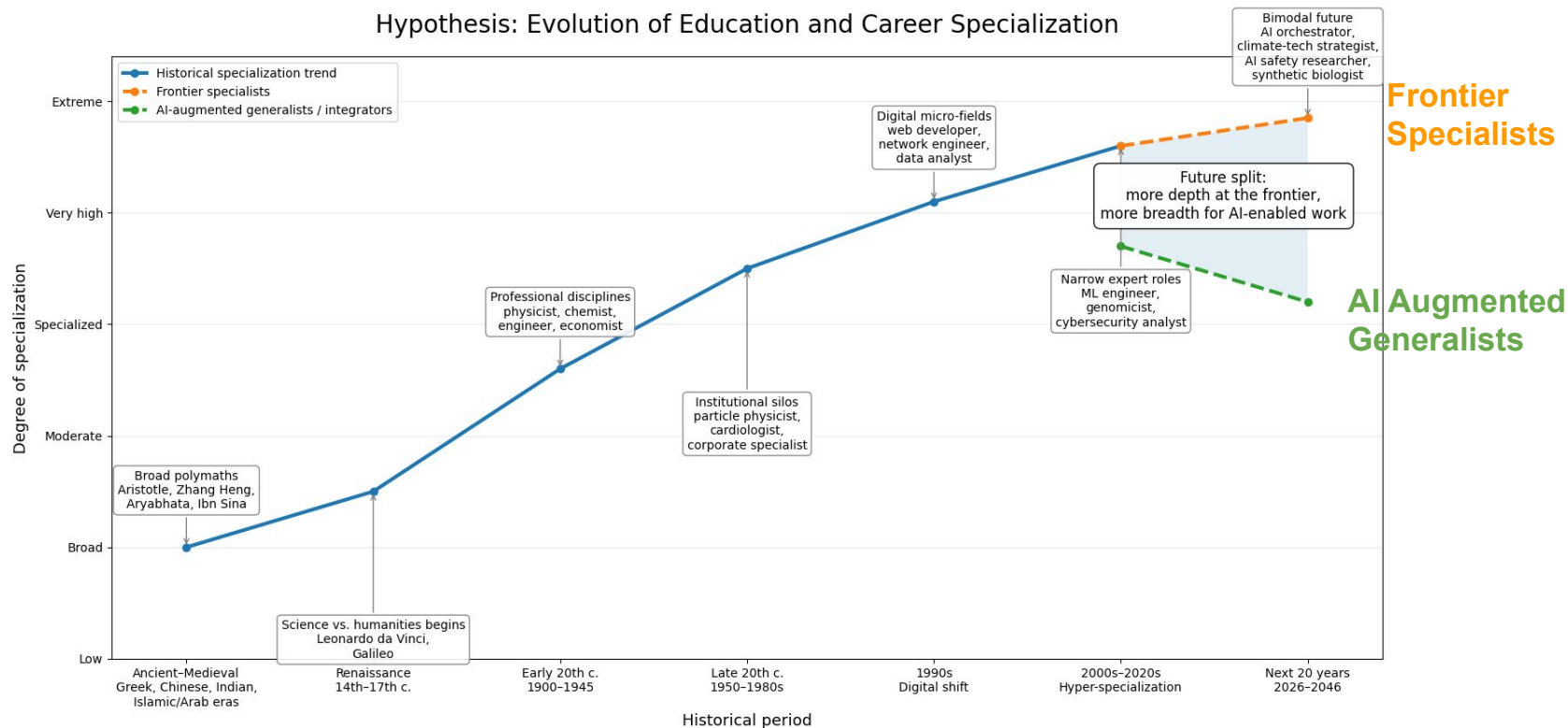
EC601

*advanced research,
real products, led by you*

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Graduate level Thesis, Project or
Company Lead R&D Project

Hypothesis: Evolution of Education and Career Specialization



Conceptual model: not a measured dataset. It illustrates a shift from broad polymathy to disciplinary specialization, then to a future split between frontier experts and AI-augmented integrators.

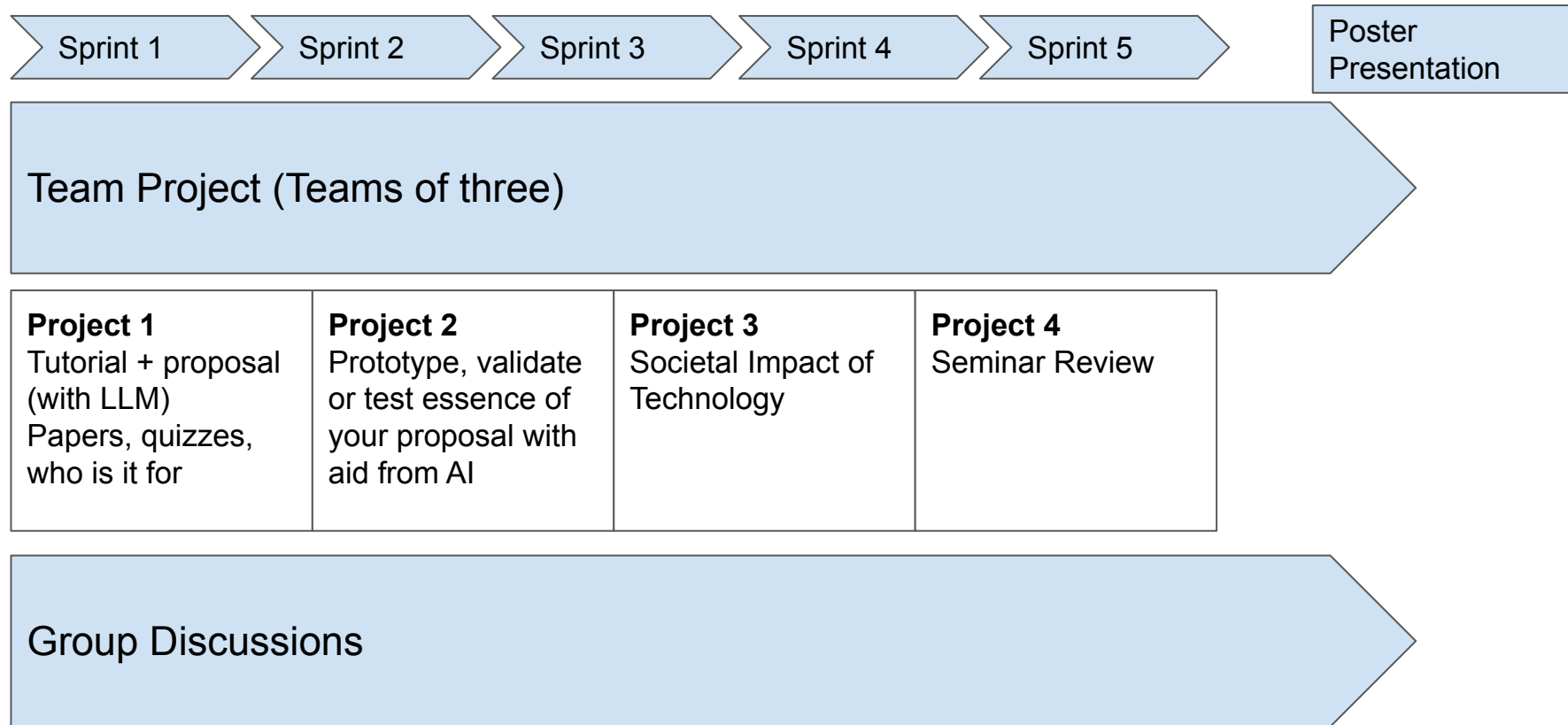
Preparing you to be a technology leader

- **Product formation:** how to take an idea and make it into a product
- **Product analysis:** competitive analysis, landscape analysis
- **Use case development:** translating use cases into system and engineering requirements, resulting in a system architecture
- **Product testing:** how to test your product
- **Concept validation:** how to validate concepts
- **After launch:** maintain, monitor and upgrade products

Class Style

- Self-learning
 - You need to research, read and train yourself with help from us.
- Utilize AI as a companion and a colleague BUT you are responsible for the outcome
- Agile Learning and development
- Class collaboration
- Very interactive and you need to present
- Always have your computer with you in class

Class Plan



Academic Misconduct (copied from Prof. Clem's email)

BU takes academic integrity very seriously. Academic misconduct is conduct by which a student misrepresents his or her academic accomplishments, or impedes other students' opportunities of being judged fairly for their academic work.

Knowingly allowing others to represent your work as their own is as serious an offense as submitting another's work as your own. More information on BU's Academic Conduct Code, with examples, may be found at

<http://www.bu.edu/academics/policies/academic-conduct-code>

Collaboration Policy (adapted from Prof. Clem)

- This is a research and project-based class. You may build on open-source code and published research — but you must reference it, explain why you used it, and follow its license. If the license does not allow use, do not use it.
- You may talk to people outside the class (engineers, researchers, potential users). Credit them as sources, just as you would a paper.
- All reports and papers must be written in your own words. Code may be shared within your team.
- Acknowledge every source — papers, code, people, and LLM chats.
- Failure to do so constitutes plagiarism and will be treated as cheating. Not sure whether something is allowed? ASK ME first — it is much easier than explaining it afterwards to the college disciplinary committee.

LLM

- You are expected to utilize AI LLM during the whole class.
- LLM is a companion or someone who works for you. This means you are responsible for the outcome
- You MUST validate every interaction
- You must include the reference to the chat in your reports
- You must include instruction to the LLM to
 - Not to make up facts
 - Not to just agree with you but to debate
 - Provide references from refereed and/or trustworthy sources
- When you are learning a new topic using AI, you must ask the LLM
 - Provide quizzes during the interaction to validate your understanding of the topic

How Classes Work

- You are expected to work on your own. Some classes will be cancelled in lieu of in person meetings with you.
- In class will be for 1 hour
- We will have open office hours for support
- We will have a teaching assistant assigned for groups
- Introduce the ambassadors

All online

- All your homeworks, documents and projects must be on GitHub
- Your GitHub account is required
- The next week, focus on becoming familiar with Github and principles of code management and agile development

Grading

Item	Item	Grade	Totals	Details
Team Grade	Sprints	20%	40%	Main criteria is effort during the sprint period.
	Final Deliverable	20%		Main criteria is poster presentation, your overall effort during the project
Individual	Project 1: Proposal	15%	60%	Main criteria is covering the main areas the report requires
	Project 2: Prototype	15%		Will be defined during user stories lecture
	Project 3: Societal Paper	10%		Will be defined during societal impact lecture. Part of the grade is for group presentation.
	Project 4: Seminar Review	10%		Pick any scientific seminar, attend it and do a thorough literature review of it.
				Attendance, in-class discussions and presentations, and engagement in the 1:1 team meetings that replace some lectures
	Participation	10%		
Total			100%	

Seminar Review

- Adapted from Prof. Pisano's Senior Design course.
- Objective: literature review of a topic and understand it well. Like Project 1, but focused on an area where people have already worked and achieved results.
- Expectations:
 - Pick a seminar, presentation, PhD defense, etc. at BU or outside BU and attend it
 - Read the related papers and work
 - Write a paper summary describing
 - Topic
 - Main contributions
 - Results
 - Your view of the work
 - What you would do as next steps
- Due date: submit before December 1, 2026 (no deadline extension).
- Resources
 - BU ENG events
 - BU CISE events
 - BU College of Computing and Data Sciences events
 - MIT Media Lab events

First Project

- Objective: Pick an area of research, topic, investigate it and put a proposal for an idea you want to pursue.
- [EC601 — Phase 1: Research & Project Areas](#)
- [The assignment description](#)
- [Survey](#)
- Create a Folder with in [The submission folder](#). The folder should be named: FirstName_LastName

Next week plan

- Wednesday 9/9/2026:
 - I will group you based on your topics
 - First 45 minutes: You will exchange ideas in groups
 - Second 45 minutes: Few groups will present their ideas
- Project 1 is due 9/12/2026

Questions

Thank you